

# Online Mass Flow Prediction in CFB Boilers

The original paper [↗](#) contains 12 sections, with 8 passages identified by our machine learning algorithms as central to this paper.

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## Paper Summary

### SUMMARY PASSAGE 1

#### Introduction

In this paper we focus on the problem of online mass flow prediction. This is an important problem since knowing mass flow is necessary for the control optimization, but measuring it directly in the CFB operational settings is difficult. Currently, the mass flow is calculated offline using relatively long time period averages.

### SUMMARY PASSAGE 2

#### A Data Mining Approach For Cfb Understanding And Control

The supercritical CFB combustion utilizes more cleanly, efficiently, and sustainable way coal, biofuels, and multifuels, but need advanced automation and control systems because of their physical peculiarities (relatively small steam volume and absence of a steam drum). Also the fact that fuel, air, and water mass flows are directly proportional to the power output of the boiler sets tight demands for the control system especially in CFB operation where huge amount of solid material exist in the furnace. When the CFB boilers are becoming larger, not only the mechanical designs but also the understanding of the process and the process conditions affecting heat transfer, flow dynamics, carbon burnout, hydraulic flows etc. have been important factors.

#### SUMMARY PASSAGE 3

### The Problem Of Obtaining Mass Flow Signal

Due to the fluctuation in the scale signal no reliable online data can be obtained from the mass flow of fuel to the boiler. The measurement system cannot be easily improved and therefore the mass flow is calculated offline using longer time period averages. The aim of the study was to remove the problematic fluctuation and fuel filling peaks from the data using signal processing methods in order to obtain online mass flow signal which could be used among others for mass flow control of fuel at the CFB-pilot.

#### SUMMARY PASSAGE 4

### Our Approach: Context-Sensitive Learning

For this study, it is assumed that the transitions from fuel feeding processes to fuel consuming processes are known. The focus will be on a learning algorithm that is able to track the mass flow given the high frequency nature of the signal and the changing fuel types for each of the processes.

#### SUMMARY PASSAGE 5

### Integration Of Domain Knowledge And Experimental Settings Metadata

The jamming of the fuel particle causes an abnormally large upward peak to the measurements that can be seen from Fig. 3. The speed of the mass change in the tank at a given time depends not only on the rpm of the feeding screw and the speed of the replenishment of the fuel in the tank, but also on the amount of the fuel in the tank. The more fuel is in the tank the more fuel gets in the screw, since the weight of the fuel at the higher levels of the tank compresses (increases the density) the fuel in the lower levels and in the screw.

#### SUMMARY PASSAGE 6

### Application

We distinguish the two types of the periods in the experiment: the consumption (fuel is only consumed) and the consumption with fuel replenishment. When one period of the CFB operating changes to another (i.e. a new portion of fuel is being added) the process of mass flow approximation and prediction starts over again, as the model of the mass flow changes. Thus, the most problematic unstable regions are the transitions intervals, when the parameters of the model change their values.

## Experimental Results

As the main points of interest are the points, where the change of the period occurs, for the analysis we took an interval of the experiment containing both types of the transition points (see Fig. 5). The computations of mass denoising were done for the delay time varying from 0 to 20 samples/seconds with step 1. In Fig. 5 and 6 the resulting approximation of mass for zero delay is shown in different scales.

## Conclusions And Further Work

A delay of less than 5 seconds allows to predict the mass flow accurately enough to be used as a reliable indicator for the CFB control system. The directions of our further work in mining CFB sensor data include (1) studying the effect of fuel feeding speed, and the effect of using different mixtures of fuels, and (2) external validation of our online mass flow prediction approach being implemented as part of the control system of the pilot CFB boiler in operational settings with different conditions. We anticipate that local reliability estimation of mass flow prediction [7] may be helpful for the domain experts.